

Eco-Procédés UGV pour les Moteurs et Energies



Influence of cutting edge preparation method on flank wear of milling inserts coated with TiN Robin test for international research (CIRP)

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Introduction

In this Robin test, we use 15 carbide inserts (Fig.1) coated with TiN, whose edges have been prepared by **abrasive blasting**, **abrasive flow machining**, **brushing**, **drugging**, **honing**, **laser machining** and **magneto-abrasive machining**.

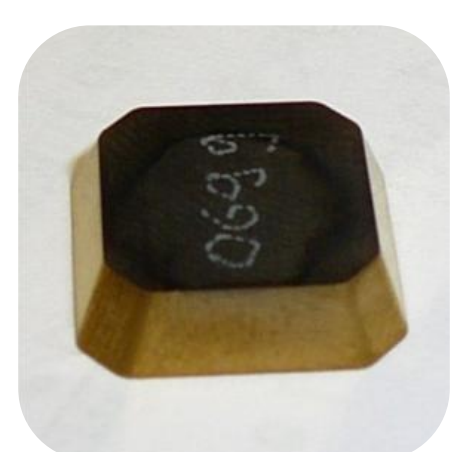


Fig.1. Insert SEAN 1203 AFTN-M14 from SECO Tools

The experiments aim to explore the relationship between **cutting edge preparation method** and the evolution of tool **flank wear VB** during down milling process on stainless steel **AISI 304L** and hardened steel **42CrMo4**. Further more, we are interested in **cutting forces** evolution, **equivalent radial angle**, **roughness** and **chip geometry** variation which correspond to the evolution of flank wear.

Process

Summary

- Measurements before tests
 - ✓ Verifying the material properties
 - ✓ Reconstructing inserts profiles
 - ✓ Analyzing roughness and r_p
 - Preparation & Design of milling procedures
 - ✓ Workpiece installation
 - ✓ NC programming
 - Perspectives
 - ✓ Online measurements
- Influence of inserts preparation on wear behavior
➤ Correlation between V_B and other parameters

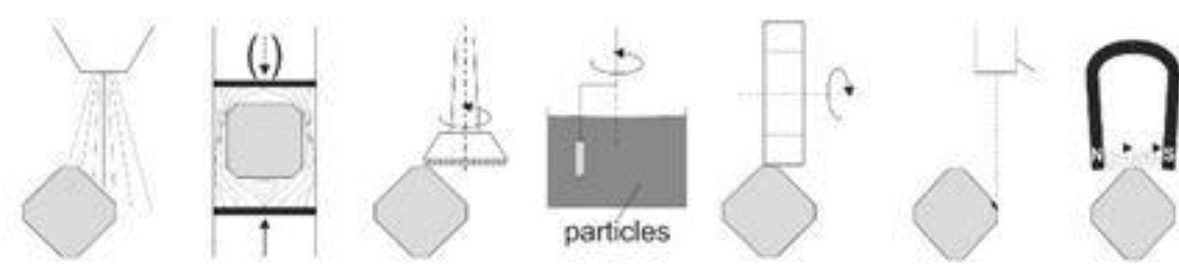


Fig.2. Inserts preparation methods

Insert N°	Pre-treatment method	Untreated (reference)	Abrasive blasting	Abrasive flow machining	Brushing	Drugging	Honing	Laser machining	Magneto-abrasive machining
0374, 0375		0429, 0430	0025, 0026	0575, 0576	0244	0699, 0700	0639, 0660	0532, 0533	

Insert & Material characterization

Insert geometry characterization

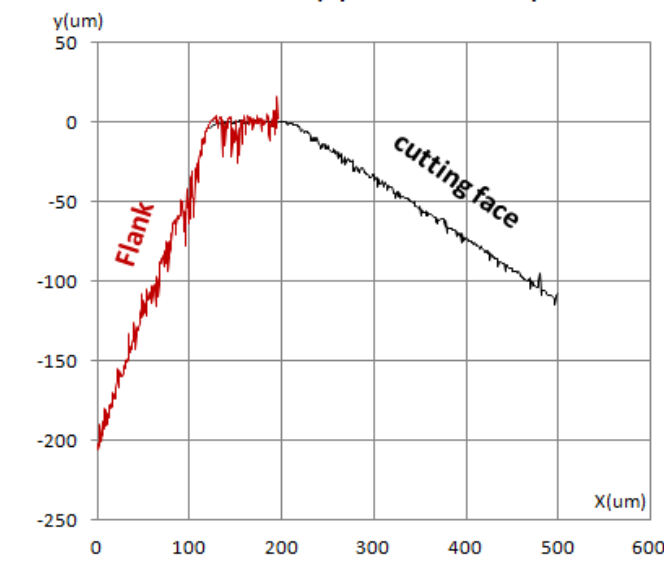


Fig.3. Combined profile of Wyko

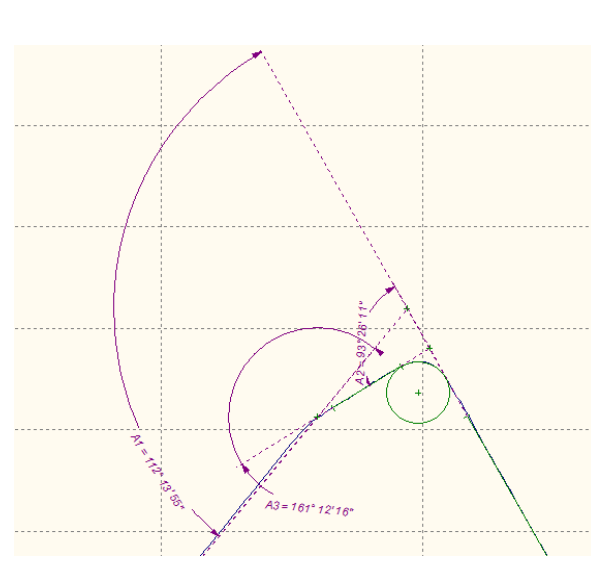


Fig.4. Profile reconstructed by SurfaScan

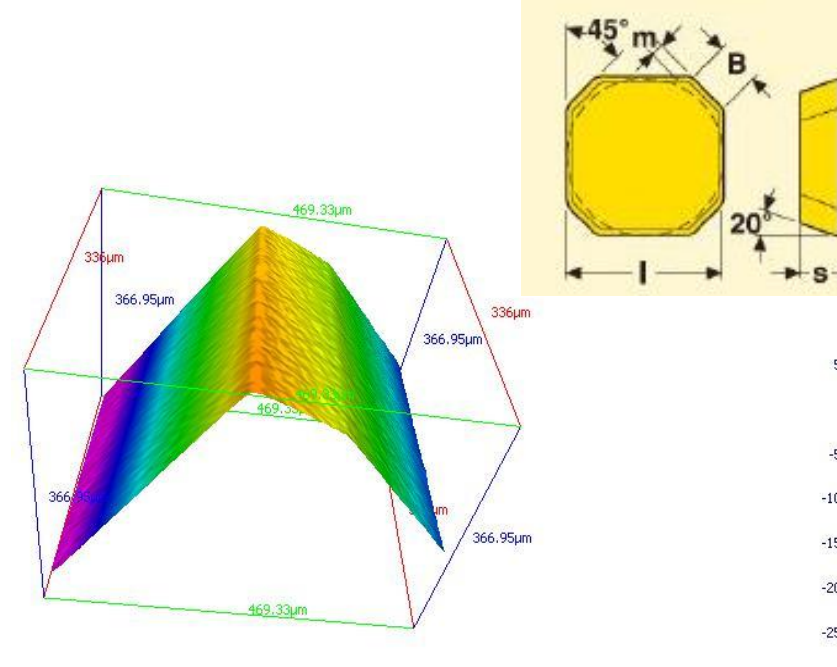


Fig.5. Profile reconstructed by SEM

Material characterization

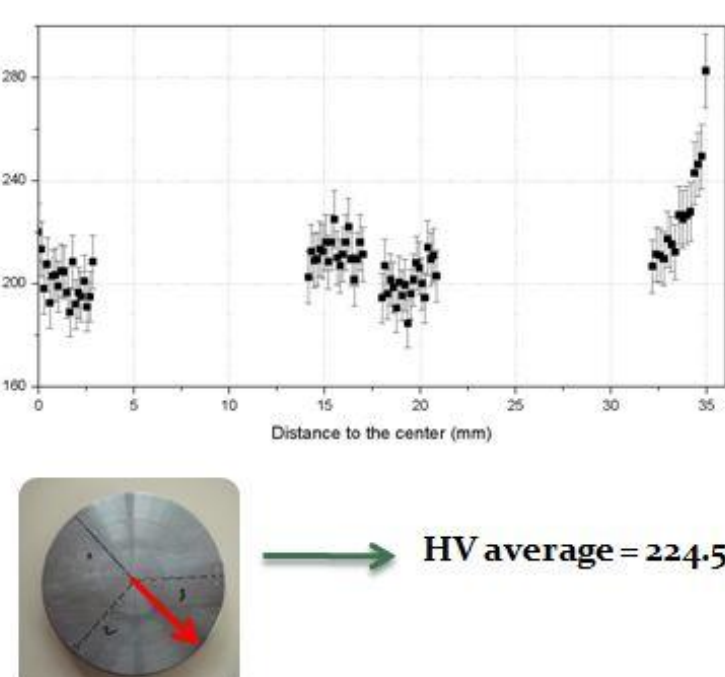
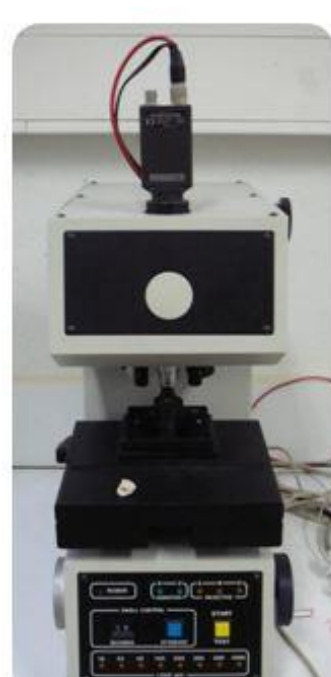
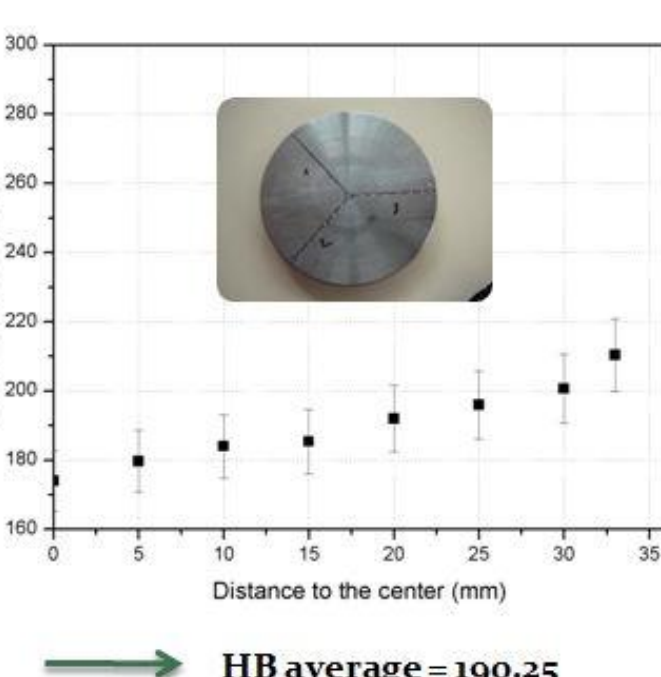


Fig.6. Macrohardness (left) and microhardness (right) of AISI 304L measured with Testor apparatus

Down milling trials

Workpiece material	42CrMo4 QT	AISI 304L
Vc [mm/min]	250	300
Process	Down milling	Down milling
Dc [mm]	50	50
Fz [mm/tooth]	0,485	0,36
a _p [mm]	3	3
a _e [mm]	4,7	3
Resulting h [mm]	0,2	0,12
Resulting l _{cu} [mm]	16	12

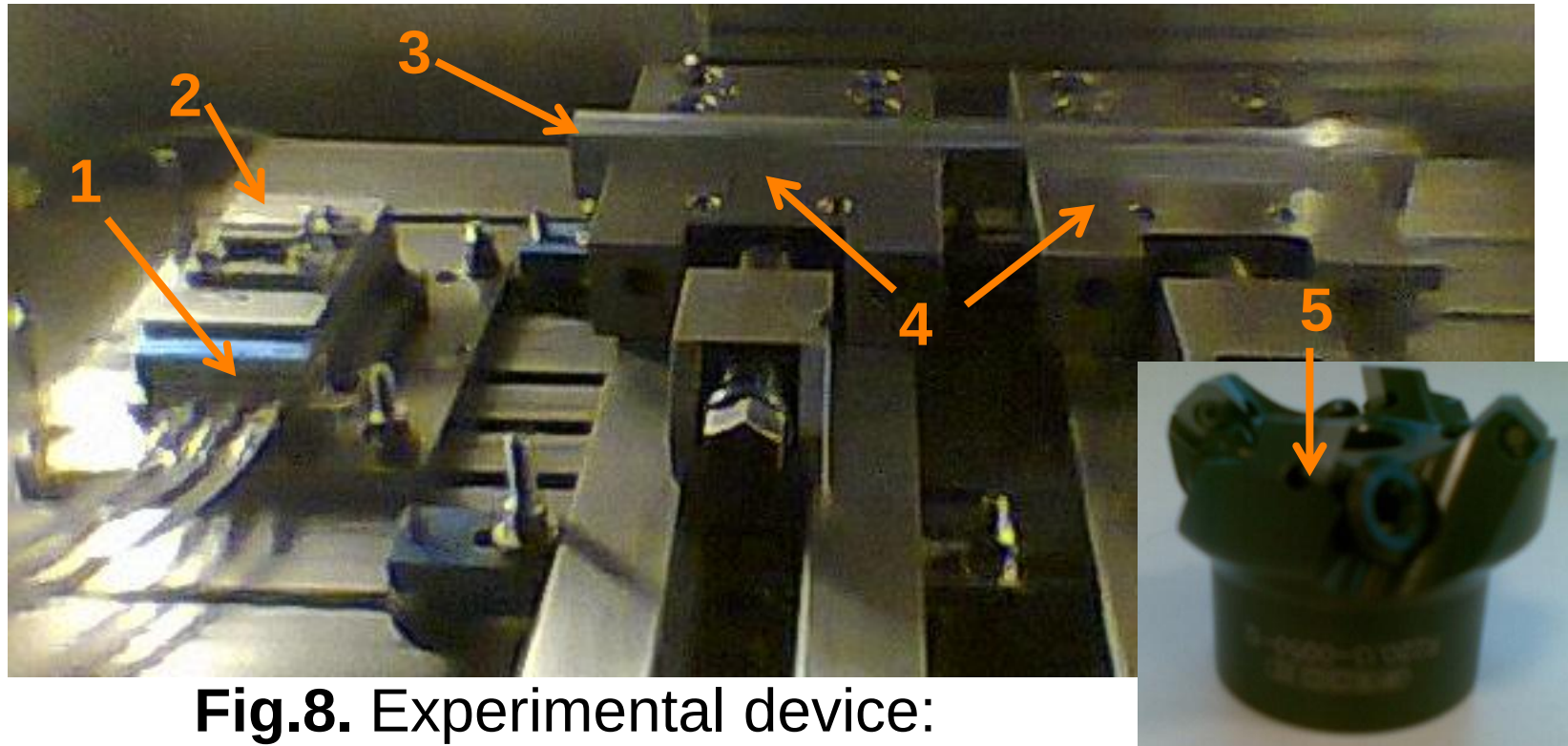


Fig.8. Experimental device:
(1)Dynamometric force transducer KISTLER 9257A
(2) plat workpiece in AISI 304L (3) main workpiece in AISI 304L (4) Clamps (6) Insert holder R220.13-0050-12 from SECO Tools

The trials are done with the experimental device described in Fig.8. The machine used is a **3 axis milling machine DMG DMC 85V** with spindle speed limit of 18000 rev/min and power limit of 25kW.

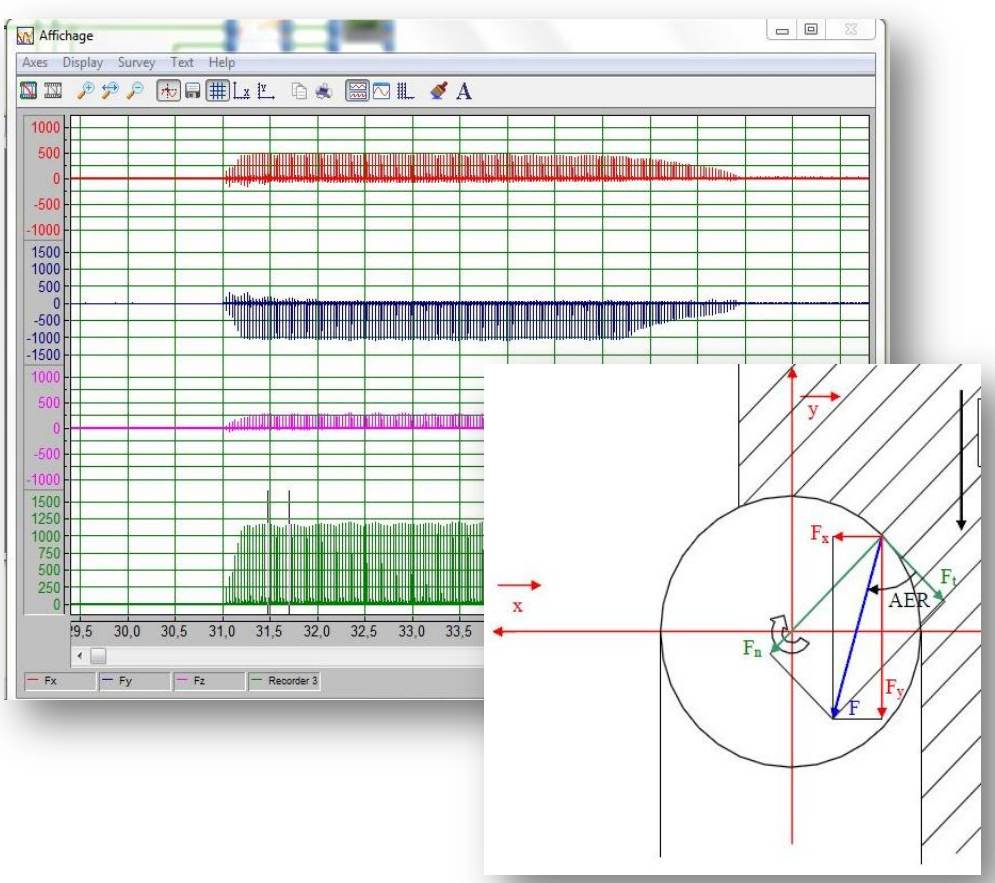


Fig.10. Cutting efforts measurement by DasyLab 11.0

Measurements includes not only V_B , but also **cutting forces**. As a consequence, after every flank wear measurement, a **force transducer** (Fig.8.(1)&(2)) and a plat workpiece are used.

Fig.7. Cutting conditions

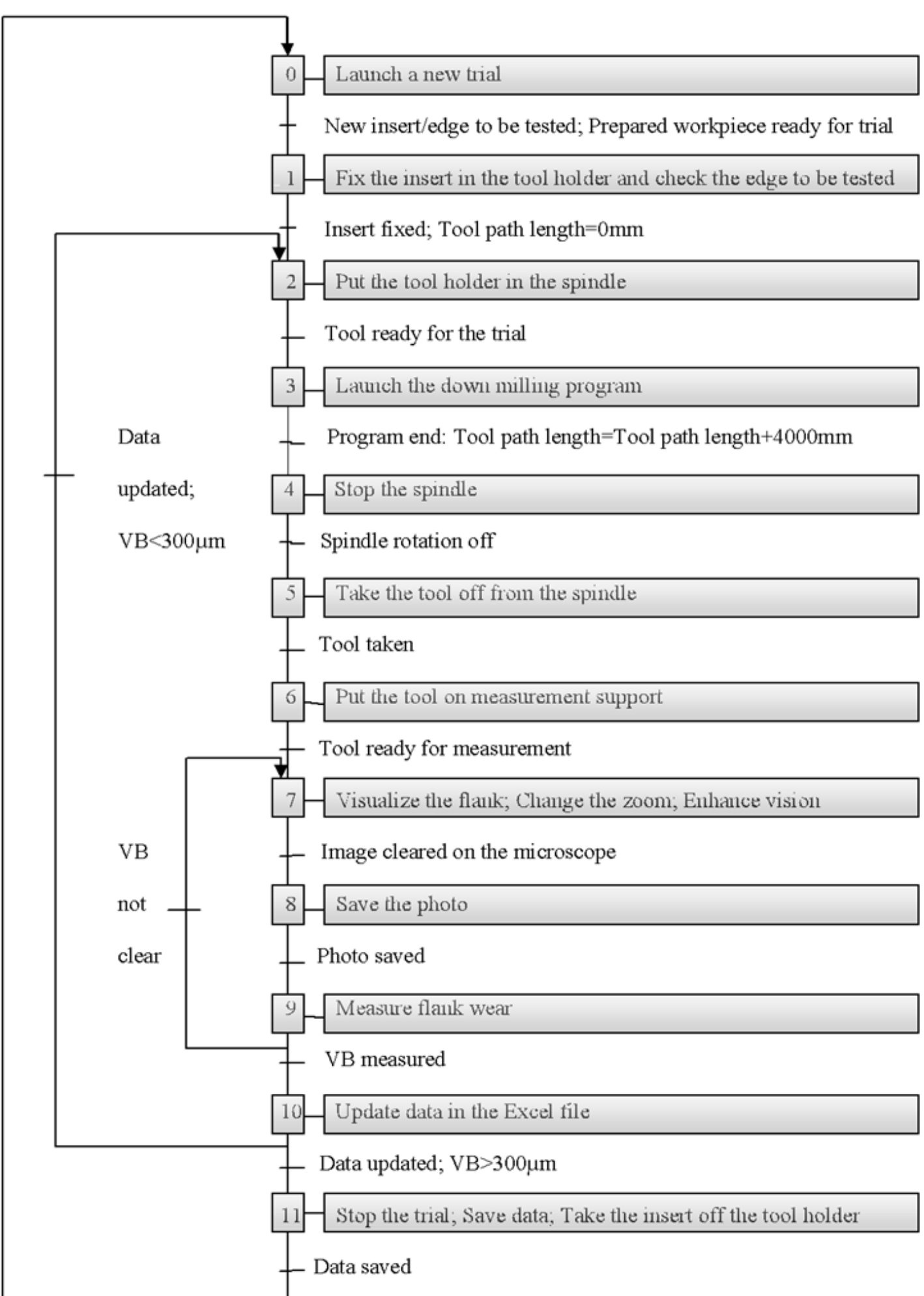


Fig.9. Trials steps

Flank wear charts



Fig.11. Digital microscope KEYENCE with High Dynamic Range function used for V_B measurement

2 edges used/insert
1 wear chart/preparation

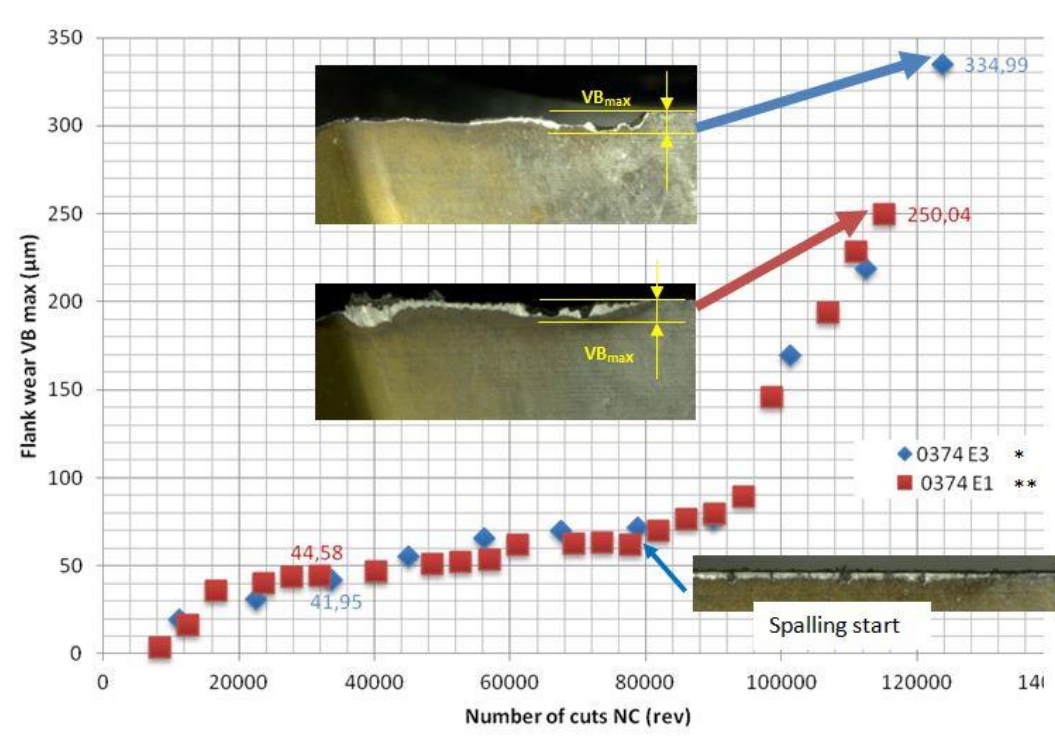


Fig.12. Flank wear chart of the insert without edge preparation down milling AISI 304L

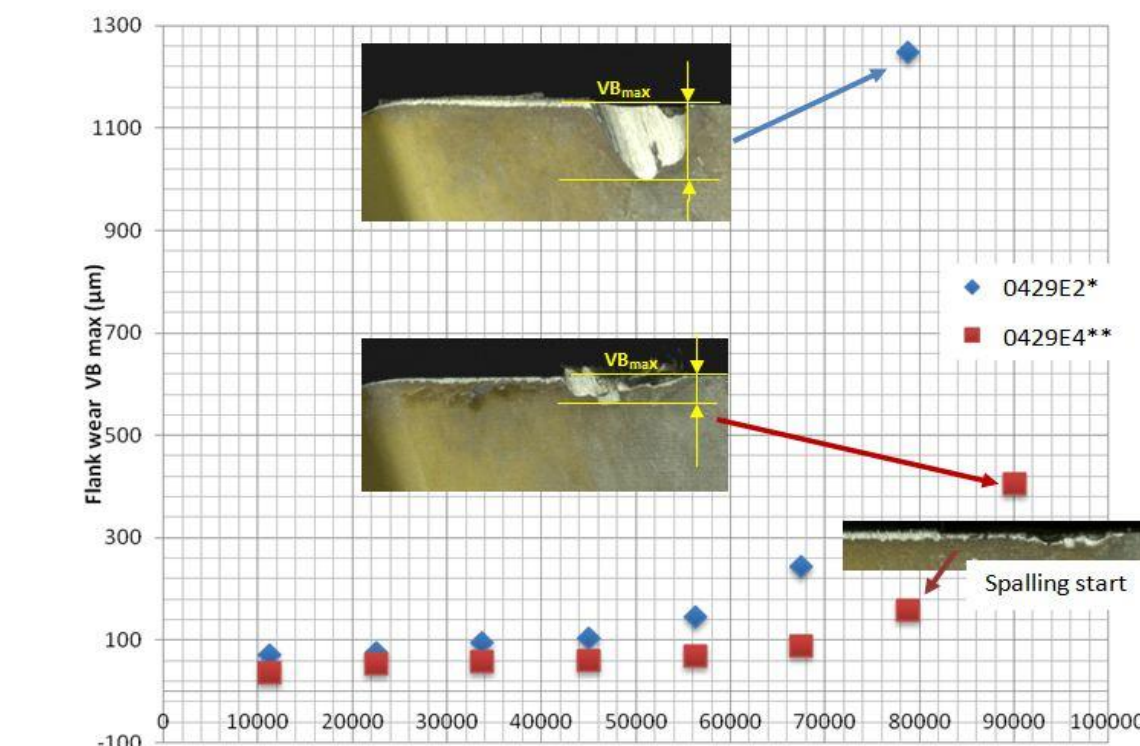


Fig.13. Flank wear chart of the insert with abrasive blasting edge preparation down milling AISI 304L

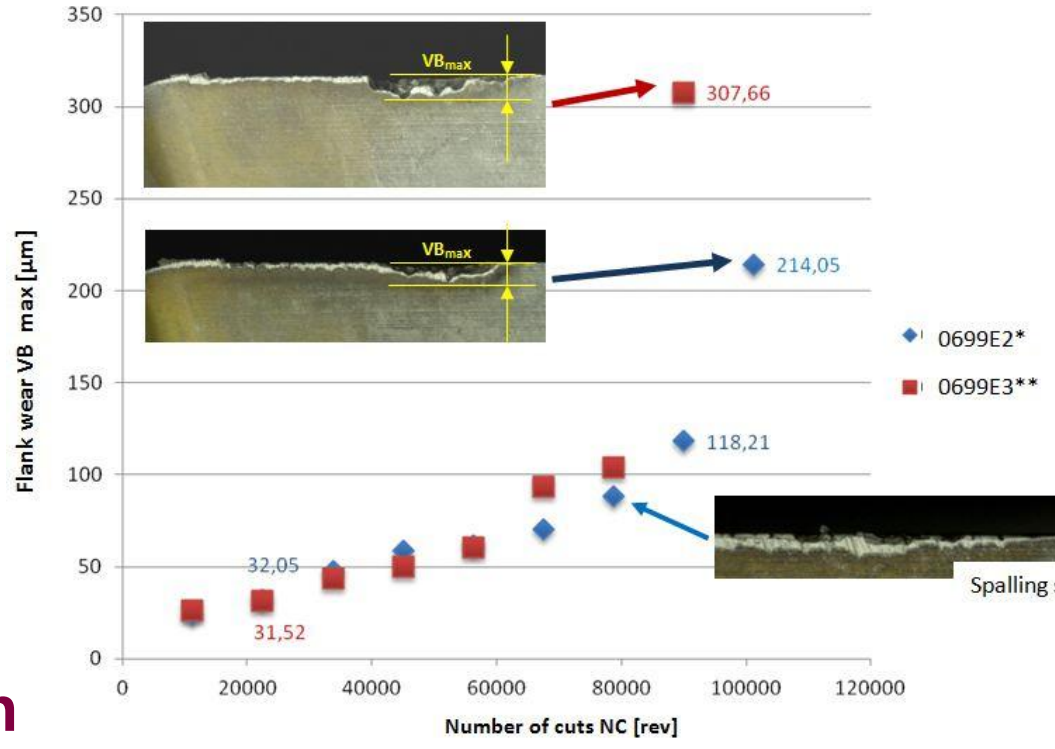


Fig.14. Flank wear chart of the insert with honing edge preparation down milling AISI 304L

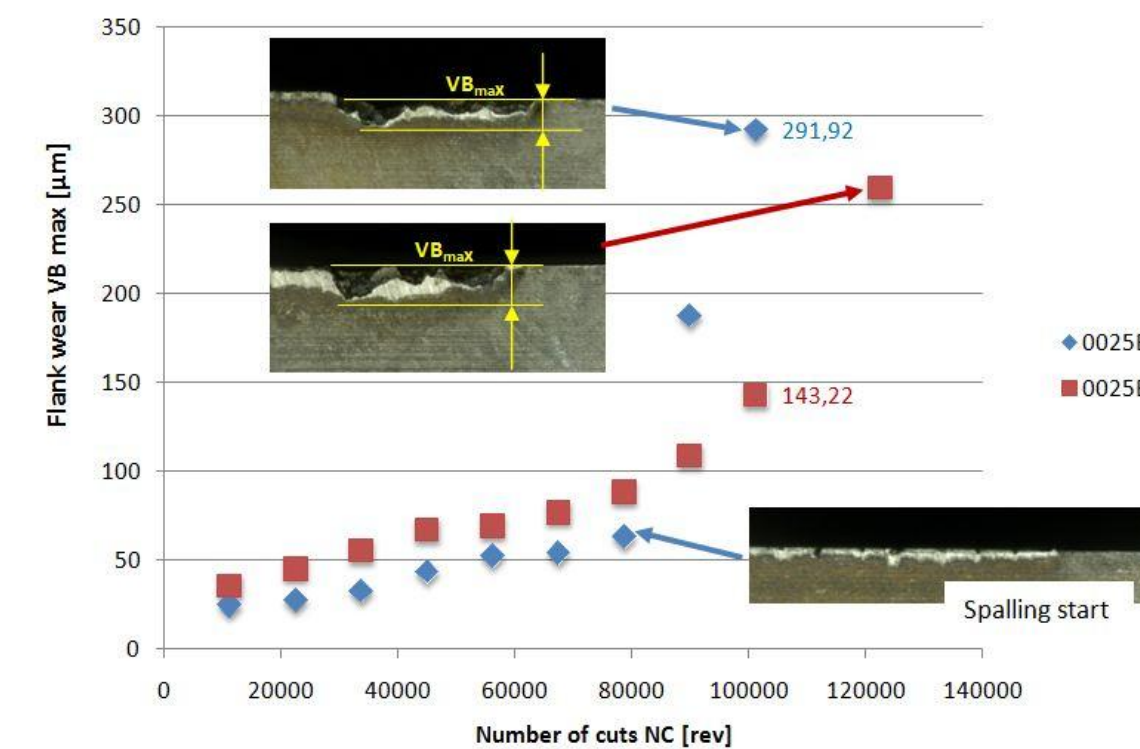


Fig.15. Flank wear chart of the insert with abrasive flow machining down milling AISI 304L

*In the current phase (April 2012), we have already finished down milling on AISI 304L by the inserts of 4 different preparations.

Edge preparation impact

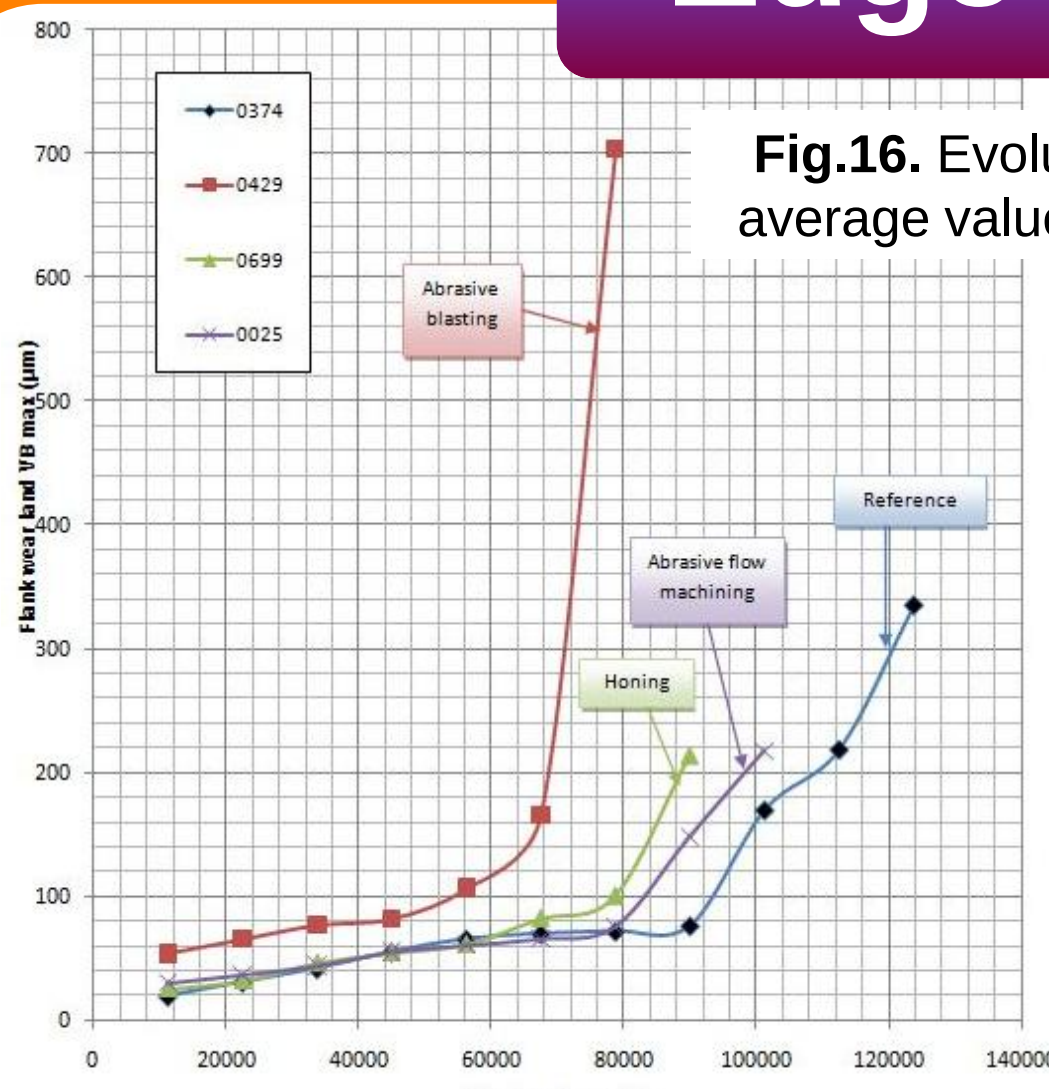


Fig.16. Evolution of average values of V_B

The comparison of tool life appears to be:
Reference > abrasive flow machining > honing > abrasive blasting.

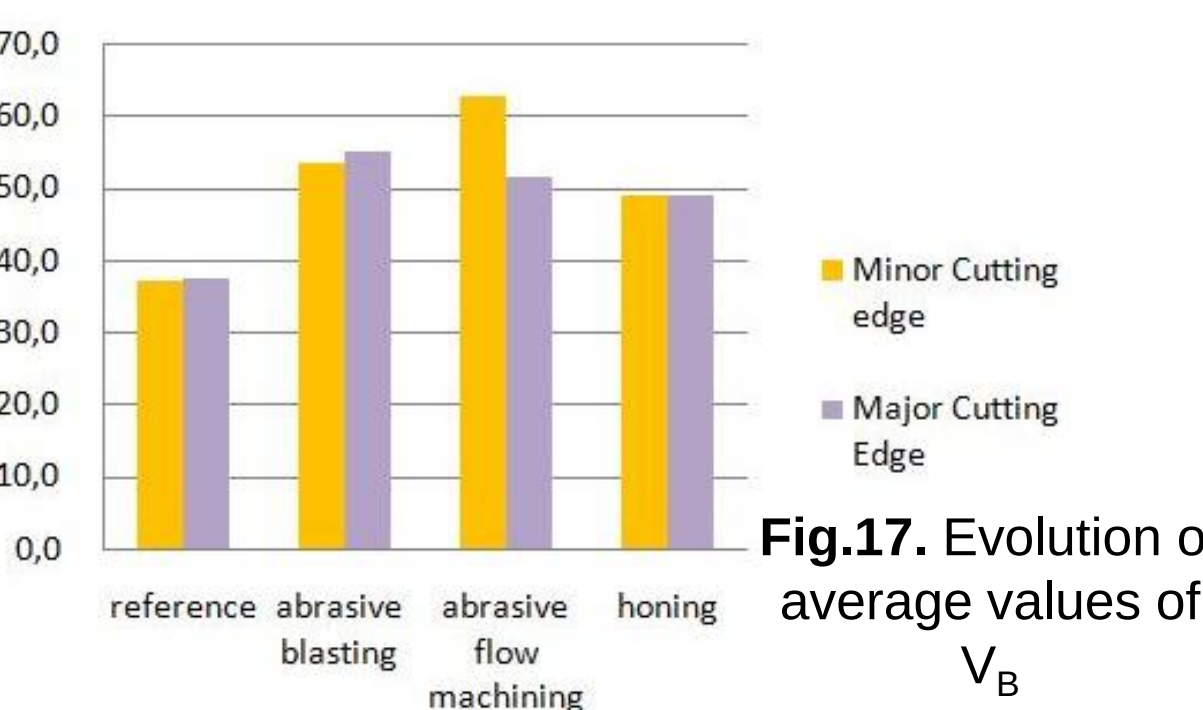


Fig.17. Evolution of average values of V_B

The main consequence of the preparation is the increment of the tiny radius between chamfer and flank, r_p (Fig.17). However, we see no evident pattern of tool life influenced by initial profile parameter, Moreover, metallurgic and mechanical aspects could affect tool life.

Bibliographie

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